

Assigned: 28th January 2025

Due: 12th February 2025 before 11.59PM

Marks: 20 marks possible

Submission Procedure:

- **Create / debug / test your assignment in Visual Studio**
- Create a **ZIP** file of your entire CareerCloud solution and upload it to **BlackBoard**.
 - o Please ensure you upload the entire solution
 - o Name your ZIP file according to the guidelines in the '1 – Sundry' slide deck
 - E.g. NET.31-5-Joe-Smith-N#####.zip
- **DO NOT EMAIL YOUR ASSIGNMENT** – The Humber email server will not deliver the assignment and you will not be notified that it was not delivered.

Description:

In this assignment you will create a REST service interface into the backend system.

Steps:

1. Add a new project to the 'CareerCloud' solution.
 - a. The new project will be of type 'ASP.NET Core Web API'. **The framework selected should be .NET 8.0 (Long Term Support).**
 - b. The name of the new project will be 'CareerCloud.WebAPI'
2. In the 'Controllers' folder that was created as part of the 'CareerCloud.WebAPI' project add a controller class for each POCO being exposed.
 - a. For example the 'ApplicantEducationPoco' class will have a corresponding controller class called 'ApplicantEducationController'.
 - b. All controller classes will have a [Route] attribute.
 - i. All route prefixes will start with 'api/careercloud/' and the type of POCO class they are exposing (e.g. applicant, company, security or system). Finally, all route prefixes will be suffixed by '/v1'.
 - c. All controller classes will inherit from the 'ControllerBase' class.
 - d. All controller classes will create an instance of their corresponding logic layer in their constructor and persist that instance in a private backing field.
3. Each controller class will have the methods for CRUD operations and an additional method to get all the records stored in the database of the corresponding POCO.
 - a. All controller methods will access data **only** through the business logic layer. Do not call methods on the repository layers directly, only the methods on the logic layer.
 - b. All controller methods will have their corresponding verb attribute [HttpGet], [HttpPost], [HttpPut] and [HttpDelete]

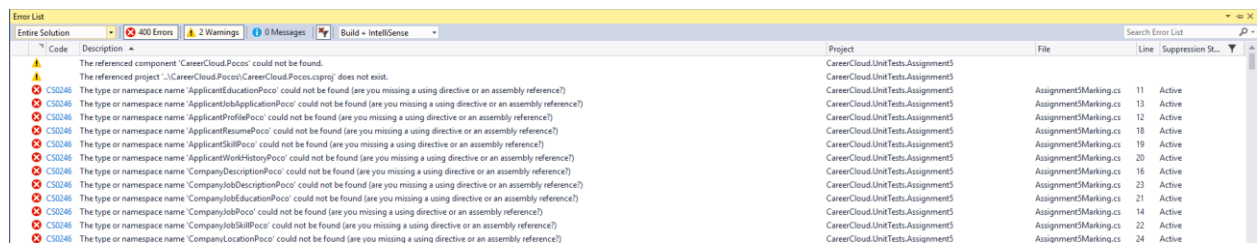
- c. All controller methods will have a [Route] attribute
 - i. The [Route] attribute will contain the POCO name without the major class prefix. For example the 'ApplicantEducationController' method GET verb GetApplicantEducation will have it's [Route] attribute as '[Route("education/{applicantEducationId}")] where the 'applicantEducationId' is the ID (GUID) of the ApplicantEducationPoco being searched for.
 - d. All controller methods will return a ActionResult type.
 - i. All controller methods will return a HTTP Okay status if the action was successfully executed.
 - ii. For methods that return a value if no corresponding record is found in the database the method should return a HTTP Not Found status.
 - e. Controller methods that return a value will have the [ProducesResponseType] attribute. The attribute will specify what the type is that will be returned from the method.
4. Install the Swashbuckle NuGet package into the 'CareerCloud.WebAPI' project
 - a. Swashbuckle is a useful tool for testing and debugging your REST projects.

Testing Your Code:

You can test your code before submitting. There is a project on GitHub located here:

<https://github.com/johnhinz/CareerCloudCore.UnitTests.Assignment5.git>.

- Clone the solution from GitHub using Visual Studio.
- Open the solution.
- Update the solution to .Net 8 and update the packages referenced if required.
- Build the test project
 - You should receive an error like the following (these errors are okay):



- Copy the test project folder into your CareerCloud solution folder.
- Add the project to your CareerCloud solution through Visual Studio
 - In your assignments solution explorer window right click and select 'Add' -> 'Existing project'
 - Navigate to the 'CareerCloud.UnitTests.Assignment5.csproj' file you copied previously
- Build the CareerCloud solution.
- Open the Test Explorer window. (Menu: Test -> Windows -> Test Explorer)

You should now be able to run the unit tests on the assignment.

You can also use Swashbuckle to test your assignment prior to running the unit tests:

<https://github.com/domaindrivendev/Swashbuckle>

Marking Scheme:

Item	Maximum Mark	Your Mark
Add the 'CareerCloud.WebAPI' project to the solution (step 1)	1	1
Create the controller methods for each POCO class (step 2)	3	3
Create the controller methods for CRUD operations (step 3)	15	15
Install the Swashbuckle Nuget package	1	1
Total	20	20